# Patrick Thangarajah

Product Design | Prototyping | Testing & Validation | Simulation

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**Hiring Team** *11 November 2026*

Planar Motor Inc.
Richmond, BC

**Mechanical Design Engineer/EIT/Specialist**

Dear Planar Motor Hiring Team,

I recently completed my MASc in Mechanical and Mechatronics Engineering at the University of Waterloo. Through my projects and work experience, I've enjoyed the practical side of engineering, especially creating a design and evaluating how well it performs. I was interested in Planar Motor's levitating movers because they combine mechanical design with motion control. The opportunity to work on equipment used in industries such as battery production and medical devices also appeals to me.

I've gained much of my practical design experience through building test equipment. During my capstone, I developed silicone and agar bladder phantoms and built the control and data acquisition interface for a pressurized test system. The system achieved stable, repeatable fill and void cycles up to 4.75 kPa. I continued working on the project with the client after my capstone, designing a PCB and writing five manufacturing and verification procedures for the electrical systems. I was also part of a UBC drone design team, where I helped design a payload drop system using SolidWorks and Onshape. Through several 3D-printed iterations and reusing components from previous designs, we reduced its weight by 50%.

During my mechanical engineering internship at a manufacturing company, I wrote work instructions and SOPs based on engineering drawings and customer specifications. My graduate research involved validating an LS-DYNA model against physical sled tests and investigating differences in restraint fit and response timing. I'm comfortable using SolidWorks, 3D printing, a lathe and a mill, and I'd also like to develop my CNC machining and welding skills.

I'm eligible to work in Canada and am happy to relocate to Richmond. I've attached my resume and transcripts for your review, and I'd appreciate the opportunity to discuss how my experience could contribute to the team.

Sincerely,

**Patrick Thangarajah**